PolicySeal

Local minimum-disclosure policy Agent

Competition Specification / Release Candidate

AMD AI DevMaster Hackathon 2026

Track 2 - Development & Local Deployment of Private AI Agents

Team: `wangjiachang`

Prepared: 2026-08-02 (UTC+8)

One source document. Different people and purposes. Only the minimum necessary disclosure.

1. Executive Summary

PolicySeal is a local enterprise disclosure-policy executor. For the same business document, it creates the minimum necessary version for a particular recipient, organization, controlled purpose, channel, classification, and policy version. Every decision is traceable to a stable document location, a content category, an action, a rule ID, human-readable policy evidence, and an immutable revision.

The product addresses a gap between basic sensitive-data detection and real disclosure decisions. A phone number may be sensitive in every context, but an invoice amount may be required by an auditor, forbidden to a competing bidder, and fully visible to internal counsel. PolicySeal evaluates that context before producing an output.

The competition build demonstrates:

- three real document paths: XLSX, DOCX, and text PDF;
- seven policy outcomes: Keep, Mask, Remove, Filter, deterministic Generalize,

Require Approval, and Block;

- a four-recipient action matrix for one supplier-comparison workbook;
- evidence-grounded, three-turn Review Copilot interaction;
- non-mutating what-if analysis and exact-binding independent approval;
- local RAG, tool selection, multi-step planning, memory, and privacy controls;
- Qwen3-8B BF16 served locally by vLLM on AMD Radeon/ROCm; and
- two reproducible optimization experiments with quality and safety evidence.

The build is intentionally fail closed. A parsing failure, unsupported embedded content, policy conflict, malformed model response, unavailable local model, or unresolved residual risk never becomes implicit permission to disclose.

2. Product Problem

Business documents are routinely shared with lawyers, auditors, suppliers, customers, incident-response partners, and regulators. Employees often create temporary copies and manually delete content. This creates two simultaneous risks:

1. under-redaction exposes personal data, credentials, commercial strategy, or

security details; and

2. over-redaction removes information that the recipient legitimately needs.

Traditional masking tools primarily ask, "Is this content sensitive?" PolicySeal asks a narrower operational question:

> What is the minimum information this recipient needs for this purpose through this channel?

The output is therefore a policy-governed disclosure version, not simply a file with every detected pattern hidden.

3. Competition Scope

3.1 P0 document support

Input	P0 behavior	Output
XLSX	Parse cells, map supplier rows/columns, evaluate recipient matrix, transform and filter	XLSX
DOCX	Parse paragraphs and tables, preserve useful structure, transform regions	DOCX
Text PDF	Extract text and positions, render source-page previews, evaluate traceable regions	Reference DOCX marked not for signature

Scanned PDFs and OCR are outside P0. A PDF page with no extractable text fails closed instead of pretending that analysis succeeded.

3.2 Synthetic scenarios

1. **Supplier comparison workbook.** Supplier names, contacts, quotes, target

prices, internal estimates, evaluation scores, negotiation comments, and approval notes are evaluated for four recipients.

2. **Security incident report.** Internal hostnames, credentials, named people,

root-cause hypotheses, and response details are transformed differently for an external response partner and a regulator.

3. **Supplier contract.** Commercial terms, bank details, contact data, and

nonstandard obligations demonstrate secure-channel output, approval, and personal-email Block.

All names, accounts, addresses, prices, systems, and events are fictional.

3.3 Explicit non-goals

The competition build does not provide OCR, arbitrary PDF layout restoration, email or IM delivery, enterprise SSO/IAM, DRM, malware scanning, policy authoring by an LLM, automated legal-compliance certification, or unattended handling of real production secrets.

4. Disclosure Context

A policy request is not valid without all of these first-class inputs:

- recipient profile;
- recipient identity and organization;
- controlled disclosure purpose;
- delivery channel;
- source classification; and
- policy version.

Purpose is selected from a controlled vocabulary rather than accepted as an untrusted free-form justification. A context change is a what-if simulation until explicitly confirmed. Confirmation creates a new immutable revision; it never mutates the prior result.

5. Hero Scenario: One File, Four Outcomes

The first review view is a four-column action matrix for the same supplier comparison workbook.

Recipient context	Minimum-disclosure behavior
Internal legal	Retain the complete authorized business context
External financial auditor	Retain evidence needed for audit; hide direct contacts and bank details
Supplier Bidder A	Retain Bidder A's own row and public requirements; filter competitors and remove internal strategy
Personal email	Block the entire disclosure before an output file is created

This matrix is the product's central distinction. Sensitivity alone does not determine the action; recipient, purpose, channel, and row ownership do.

6. Decision Model

Every policy decision contains:

- stable region ID and source location;
- content category;

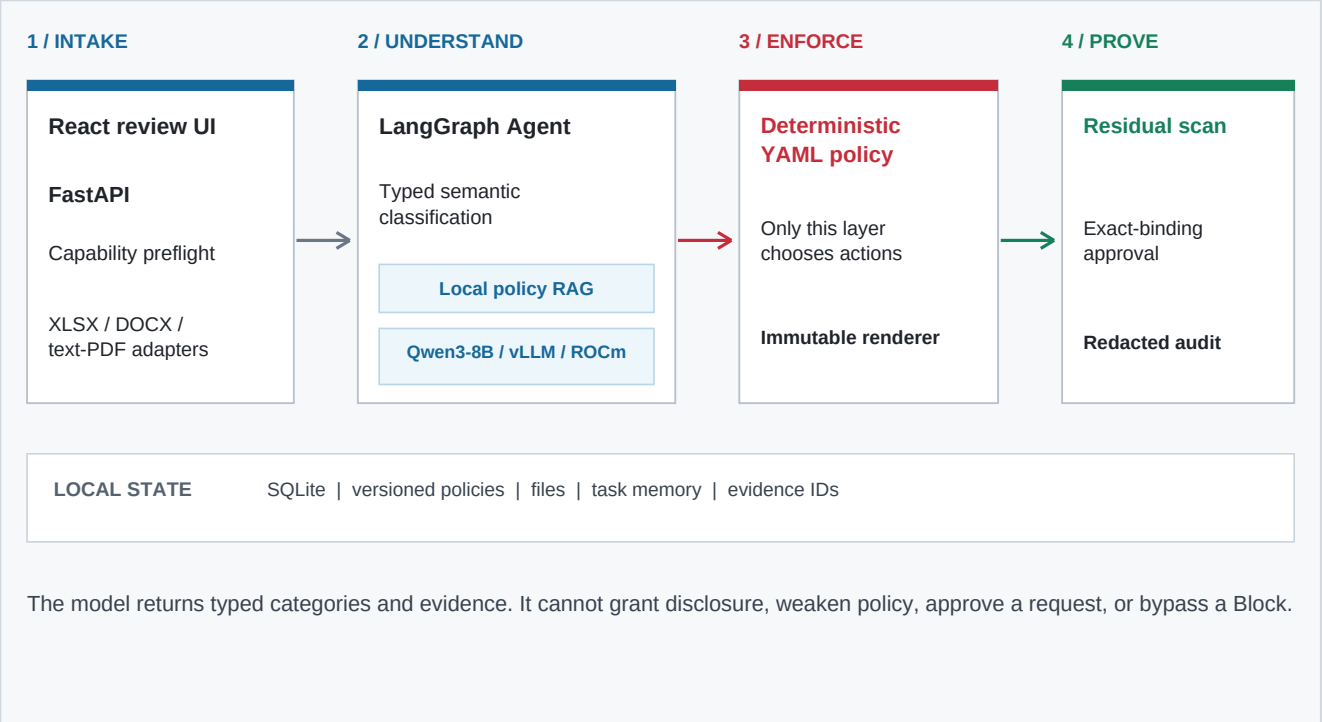
- final action;
- rule ID and reason code;
- local evidence IDs;
- policy version and policy SHA-256;
- deterministic or fail-closed origin; and
- optional deterministic transform definition.

The action precedence is:

```
Block > Require Approval > Transform > Keep
```

When transforms conflict, PolicySeal uses the intersection of information that can be proven safe. If conservativeness cannot be proven, the result requires approval. A model prediction never receives higher precedence than policy.

7. System Architecture



7.1 Layer boundaries

- The domain layer owns immutable task, revision, region, decision, approval,

and audit concepts.

- The policy layer is deterministic and side-effect free.
- Document libraries, persistence, model HTTP, and FastAPI are adapters.
- The application layer coordinates stages and creates auditable failure

decisions when output safety cannot be proven.

- The web application consumes typed APIs; it does not calculate policy.

8. Multi-Step Analysis Workflow

Each formal revision moves through named stages:

1. validate source identity and disclosure context;

2. preflight document capabilities;
3. parse stable regions;
4. apply deterministic preclassification;
5. retrieve local policy evidence for uncertain semantics;
6. classify uncertain regions through two matching local-model passes;
7. validate typed category and evidence output;
8. execute deterministic policy and resolve conflicts;
9. render a new output revision;
10. reopen the file and validate its structure;
11. scan the generated output for residual risk;
12. mark the revision Exportable, Awaiting Approval, Blocked, or Processing

Failed; and

13. append redacted audit evidence.

Unsupported formulas, embedded objects, PDF attachments, and unextractable PDF pages create stable system regions and Block decisions with rule evidence. A residual hit deletes the candidate output and creates a non-waivable Block.

9. Local AI Agent

9.1 Semantic classification graph

The LangGraph classification graph uses these nodes:

```
deterministic_preclassification
-> local_model_call when needed
-> typed_validation_and_evidence
-> human_interruption when unresolved
-> finish
```

High-confidence structured patterns skip the model. Uncertain regions are grouped by file type, limited to eight stable IDs per request, and classified twice. The model may return only a fixed content category and local evidence IDs. Pydantic validation, category-evidence compatibility, and repeated-output agreement determine whether the result can proceed.

If the local model is unavailable, unresolved semantics become Require Approval with `PS-MODEL-OFFLINE-APPROVAL`. There is no public inference fallback.

9.2 Review Copilot graph

Review Copilot is task-scoped and supports three controlled intents:

- explain a decision using its rule and evidence;
- run a non-mutating what-if simulation; and
- draft an exception request that can be submitted for independent approval.

The graph loads the exact task and revision, classifies controlled intent, invokes only the allowed tool, and stores a redacted checkpoint. It has no tool that edits policy, approves a request, changes a decision in place, or bypasses Block.

9.3 Local RAG

Human-readable policy sources cover disclosure principles, XLSX supplier handling, incident response, and contract handling. The retrieval index contains more than 40 citable chunks. Retrieval explains definitions and policy evidence; vector similarity never decides access.

The frozen retrieval evaluation contains 20 questions. The measured result is 20/20 expected evidence hits at Recall@3.

10. Document Transformations

10.1 XLSX

The workbook adapter discovers headers, row ownership, metadata, and stable cell locations. It supports cell masking/removal, row filtering, deterministic numeric generalization, and approval-bound withholding while preserving workbook structure. Formula evaluation is outside P0 and is blocked during preflight.

10.2 DOCX

The DOCX adapter processes paragraphs and table cells, strips unsupported custom XML, creates stable task-local aliases for people and hosts, removes approval- bound content from the external candidate, and reopens the resulting package. Credential residuals are non-waivable.

10.3 Text PDF

The PDF adapter extracts positioned text and produces nonblank source-page previews. The output is a traceable DOCX reference draft. Exact nonstandard obligations may remain in an approval-only review draft; other unapproved content is withheld. The output is visibly labeled `Reference Draft - Not for Signature`.

11. Approval, Export, And Audit

- A requester may submit approval work but cannot approve the same revision.
- A reviewer may approve or reject but cannot edit the generated output.
- Approval binds to the source hash, output hash, context, policy version, and

revision.

- An approved business download consumes that approval once.
- Evidence export is separate from business output.
- Blocked revisions cannot enter approval and cannot export.
- Audit records use controlled metadata and keyed HMACs when value identity must

be compared. They omit full source text, raw secrets, credentials, reversible masks, and model hidden reasoning.

12. Operational Interface

The local React interface is optimized for repeated review work rather than a marketing flow. It provides:

- task creation with explicit context fields;
- named local-analysis stages;
- the XLSX four-recipient matrix as the primary first view;
- synchronized source/result diff and rule evidence;
- PDF source-page previews and reference-output review;
- requester/reviewer role simulation clearly labeled as synthetic;
- what-if and formal revision controls;
- approval and one-time business export;
- a separate evidence package; and
- task-local Review Copilot below direct evidence.

The browser gate checks 1440x1000 desktop and 390x844 mobile layouts for horizontal overflow, clipped buttons, incoherent overlap, blank previews, and post-load movement.

13. Privacy And Threat Controls

Risk	Control
Public model disclosure	Localhost-only vLLM; public and credentialed model URLs are rejected
Model grants access	Model cannot emit an action; deterministic policy owns the result
Malformed or inconsistent model output	Typed validation and two-pass agreement; otherwise approval
Unsupported document content	Preflight Block with a stable system decision
Secret copied into logs	Sanitized structured audit and redacted checkpoints
Unsafe generated file	Reopen validation and residual scan before export
Approval replay	Exact binding and one-time consumption
What-if accidentally becomes formal	Simulations are non-downloadable until a new revision is confirmed
Golden answers fit to implementation	Frozen independent fixtures and hashes; acceptance compares exact expected outcomes

14. Evaluation Package

The repository freezes:

- 30 end-to-end scenarios;
- 60 semantic snippets;
- 20 evidence-retrieval questions; and
- 6 three-turn Copilot dialogues.

The scenario distribution is 12 Safe, 8 Require Approval, 8 Block, and 2 Processing Failed. Two scenarios deliberately inject a model outage and a residual-output hit to make failure behavior reproducible. They are not normal runtime claims.

Current final-code gates:

Evidence	Result
Frozen scenarios	30/30
High-risk leakage	0
Rule/evidence coverage	100% for asserted decisions
Python tests	179 passed
Frontend unit tests	11 passed
Browser journeys	4 passed locally and on Radeon
Retrieval Recall@3	20/20
Copilot continuity	18/18 turns

The live Qwen3-8B evaluation on all 60 semantic snippets measured macro-F1 `0.948485`, accuracy `0.933333`, one fail-closed human-review outcome, p50 request latency `8,894.77 ms`, p95 `11,318.75 ms`, and throughput `0.411368` regions per second.

15. Radeon Runtime

The verified environment is:

Component	Measured value
Platform	Ubuntu 24.04.4 LTS
GPU label	AMD Radeon Graphics
GPU architecture	`gfx1100`

Component	Measured value
Reported VRAM	`51,522,830,336` bytes
ROCm	`7.2.1`
HIP	`7.2.53211-e1a6bc5663`
Python	`3.12.3`
PyTorch	`2.9.1+gitff65f5b`
vLLM	`0.16.1.dev0+g89a77b108.d20260317`
Transformers	`4.57.6`
Model	Qwen3-8B, BF16, non-thinking mode
Attention	vLLM ROCm Triton attention
Model endpoint	`127.0.0.1:8000`
Maximum context	8,192 tokens
Maximum sequences	8
GPU memory setting	0.46
Measured vLLM cold start	34,150 ms

The platform reports a generic GPU name, so this submission does not infer an unverified retail product model. The eight critical model files are SHA-256 verified; their combined identity is `18c9bc524cc692247b4cc0c82c76260dd6539e29ab7af8e62313738624ec45b1`.

16. Radeon Optimization Experiments

The official report is bound to source commit `522ef6bcb34c41618460b66c5ca4214f849c65b5`. It uses 10 warm repetitions per strategy and two consistency passes. Raw records include latency, model request and region counts, tokens, GPU utilization, VRAM, output hashes, and quality.

16.1 Experiment A: stable-ID batching

Metric	Per-region calls	Stable-ID batch size 8
Warm p50	24,989.73 ms	22,264.37 ms
Warm p95	24,997.20 ms	22,312.93 ms
Region throughput	0.360145/s	0.404114/s
Model requests across warm runs	180	60
Peak VRAM	23,216,590,848 B	23,216,590,848 B
Macro-F1	0.629630	1.000000
Leakage	0	0

Warm p95 improved by `10.738277%`, below the approved `20%` target. The result is therefore `not_met`. The optimized configuration is still preferred because it reduced requests, improved throughput, preserved zero leakage, and passed the quality gate while the per-region baseline over-removed one sample and sent one sample to review.

16.2 Experiment B: deterministic short circuit

Metric	All regions through model	Deterministic short circuit
Model regions across warm runs	300	90
Warm p50	74,510.84 ms	22,257.27 ms
Warm p95	74,517.27 ms	22,288.29 ms
End-to-end region throughput	0.402635/s	1.347738/s
Mean GPU utilization	99.786147%	98.812857%
Macro-F1	1.000000	1.000000
Leakage	0	0

The optimized path reduced model-region count by exactly `70.00%`, meeting its target, and reduced warm p95 by approximately `70.09%` without changing accepted quality. The result validates the product architecture: deterministic rules are both the security authority and the largest measured performance optimization.

17. Reproduction

17.1 Local functional gate

```
make install
cp .env.example .env
make check
make install-e2e-browser
POLICYSEAL_E2E_SKIP_BROWSER_INSTALL=1 make e2e
make validate-evals
```

17.2 Radeon gate

```
ssh policyseal-radeon
cd /workspace/PolicySeal
bash scripts/bootstrap-cloud.sh
bash ops/radeon/download-model.sh
./ops/radeon/start-services.sh
./ops/radeon/health-check.sh

source ops/radeon/policyseal-env.sh
make check
.venv/bin/python scripts/evaluate_semantic_model.py \
--output artifacts/benchmarks/semantic-evaluation.json
make benchmark
```

Generated files and caches stay under `/workspace`. Before an instance is destroyed, source, frozen goldens, raw benchmark JSON, and required artifacts must be synchronized back to the repository workspace.

18. Track 2 Evidence Map

Judging area	PolicySeal evidence
Task definition and originality	Context-dependent minimum disclosure for the same document, not uniform masking
Core Agent capability	Tool selection, multi-step LangGraph flow, local RAG, typed model use, approval, and audit
Multi-turn interaction	Six frozen three-turn dialogues with explanation, what-if, and exception work
Local Radeon inference	Qwen3-8B BF16 through localhost vLLM on ROCm 7.2.1
Radeon optimization	Stable-ID batching and deterministic short-circuit comparisons with raw quality evidence

19. Release Position

PolicySeal is a competition prototype with strong safety boundaries and reproducible synthetic evidence. It demonstrates how a private AI Agent can help interpret uncertain language while keeping deterministic enterprise policy in control. It is not a production DLP replacement or a legal-compliance product.

The intended final takeaway is simple:

One source document. Different people and purposes. Only the minimum necessary disclosure, decided locally and explained line by line.